



Pape Magatte Ndiaye DIENE

SOFTWARE ENGINEER | AI & FULL-STACK DEVELOPMENT



Portfolio : <https://magattediene.netlify.app>

CONTACT

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Github

<https://github.com/MagatteDiene>

TECH SKILLS

Languages & Frameworks :

- PHP (Laravel), JavaScript (React, AngularJS), Python (FastAPI, Flask), Java, HTML/CSS, C#

AI / Machine Learning :

- LangChain, RAG, RAGAS, Machine Learning, LLM, Groq API

Databases :

- MySQL, PostgreSQL, ChromaDB

Other:

- API REST/GraphQL, Git/GitHub, Postman, UML Modeling

LANGUAGES

- French
- English



PROFILE

TECHNOLOGY ENGINEER IN COMPUTER SCIENCE, GRADUATE OF THE POLYTECHNIC SCHOOL OF DAKAR (ESP), WITH OVER 2 YEARS OF EXPERIENCE IN WEB APPLICATION DEVELOPMENT, RAG ARCHITECTURES, AND ARTIFICIAL INTELLIGENCE. FULL-STACK DEVELOPER AT HTSOFT SENEGAL, PASSIONATE ABOUT DESIGNING INNOVATIVE SOFTWARE SOLUTIONS WITH STRONG BUSINESS IMPACT.



EDUCATION

- TECHNOLOGY ENGINEERING DEGREE, Computer Science** 2024 - 2026
Polytechnic School of Dakar (ESP), Dakar
- HIGHER TECHNOLOGY DIPLOMA, Computer Science** 2022 - 2024
Polytechnic School of Dakar (ESP), Dakar
- Baccalaureate - Science Series (S2)** 2021 - 2022
Lycée Sergent Malamine Camara



PROFESSIONAL EXPERIENCE

Fullstack Developer – Backend Focus

HTSOFT SENEGAL

November 2024 - Present

- Designed and developed functional modules for several production business applications (procurement-logistics-distribution ERP, HR information system, real estate management)
- Integrated artificial intelligence solutions into existing applications, including an automated CV screening module
- Contributed to UML modeling and technical architecture of several applications, from functional design to deployment

Web Development Intern

HTSOFT SENEGAL

May 2024 - November 2024

- Designed and built an online seminar management platform from scratch using Laravel, from data modeling to delivery
- Led this project independently as part of the final internship required for the Higher Technology Diploma in Computer Science



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CERTIFICATIONS

Elements of AI

University of Helsinki & MinnaLearn

Google AI Professional Certificate

Google

Google AI Essentials Specialization

Google

Introduction to Generative AI for Software Development

DeepLearning.AI

Foundational C# with Microsoft

Microsoft

Cisco Networking Basics

Cisco Networking Academy

Cisco Networking Devices and Initial Configuration

Cisco Networking Academy

OTHER TRAINING

Intensive English Session (Level 3)

CPLA — Centre de Perfectionnement en Langue Anglaise

July 1st, 2026
September 30th, 2026

PROJECTS

SamaVoie — Thesis Project

Hybrid RAG system for academic guidance in Senegal: dense retrieval (BGE-M3) + BM25, RRF fusion, cross-encoder reranking, HyDE, and multi-turn query reformulation. Conversational assistant (Kali) + program recommendation engine.

Technologies: Python, LangChain, ChromaDB, FastAPI, Groq API, React

SIRH (Human Resources Information System)

Contributed to the front-end and back-end development of the HR platform, integrated an AI module for CV pre-screening, developed Laravel APIs, and deployed the application.

Technologies: React, Laravel, Postman, PostgreSQL, OpenAI API

SALAMA (Procurement, Logistics and Distribution Management ERP)

Web application managing a company's full procurement-transport-distribution cycle: tenders and supplier evaluation, transport offer validation workflow, fleet management (vehicles/drivers/shipments by tonnage), warehouses and delivery notes, network of points of sale by zone/province, and tracking of donor-funded programs. Integrates Sage X3, a GraphQL API, Excel/PDF exports, and a roles/permissions system.

Technologies: Laravel, React, Postman, PostgreSQL

GESTIMMO (Real Estate Management ERP)

Developed real estate and financial management modules, generated PDF documents, migrated data, and supported users.

Technologies: Laravel, Blade, AngularJS, Postman, PostgreSQL

N-BaloT - Network Intrusion Classification (Machine Learning)

Designed an IoT intrusion detection system based on machine learning, experimenting with several algorithms (Random Forest, SVM, KNN) and deploying a web interface.

Technologies: Python (Flask)

TamTam FinConnect

Full development of the responsive landing page, design and improvement of user modules, project owner and investor dashboards, and management of opportunity lists. This project strengthened my skills in UI/UX, front-end development, and full system design.

Technologies: HTML, CSS/Tailwind, PHP (Laravel)